



Southeast University

Department of Computer Science and Engineering (CSE)

School of Sciences and Engineering

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LAB REPORT NO: 01

Course Title: Operating Systems Lab

Course Code: CSE362.2

Batch: 65

Lab Experiment Name: Implementing fundamental Linux command.

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Lab Report Status

Marks:

Comments:.....

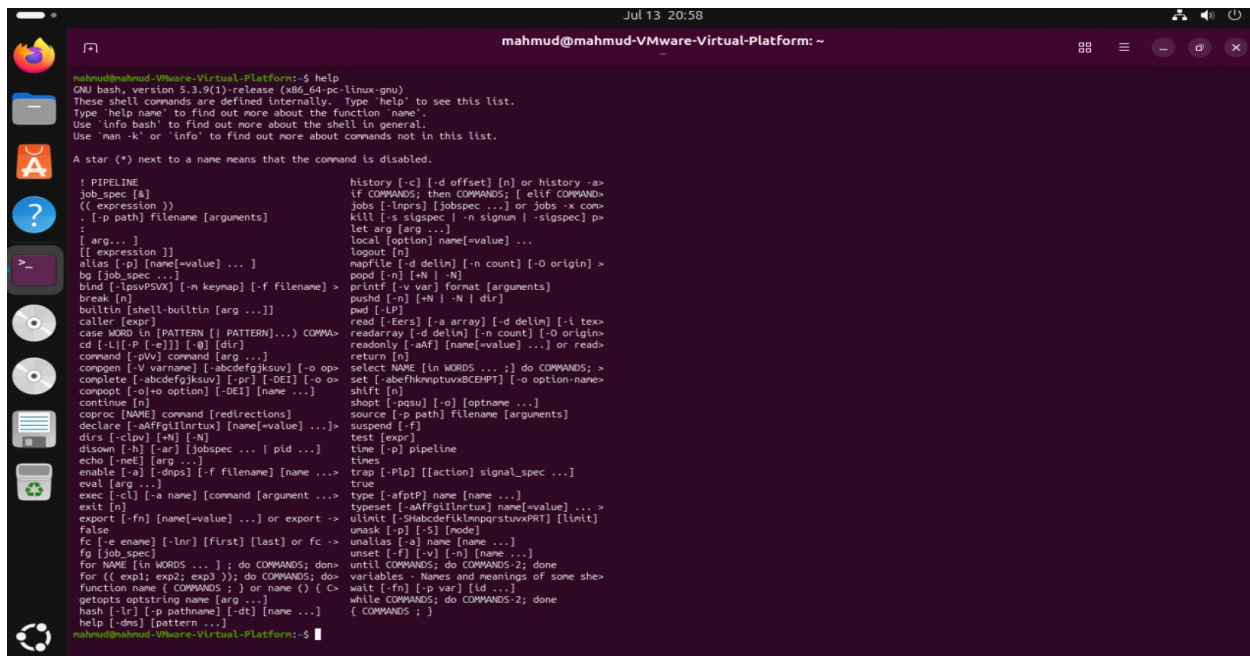
Signature:.....

Date:.....

Command 1: help

Description: Display the list of all Linux commands.

Screenshot:



```
mahmud@mahmud-VMware-Virtual-Platform:~$ help
GNU bash, version 5.3.9(1)-release (x86_64-pc-linux-gnu)
These shell commands are defined internally. Type 'help' to see this list.
Type 'help name' to find out more about the function 'name'.
Use 'info bash' to find out more about the shell in general.
Use 'man -k' or 'info' to find out more about commands not in this list.

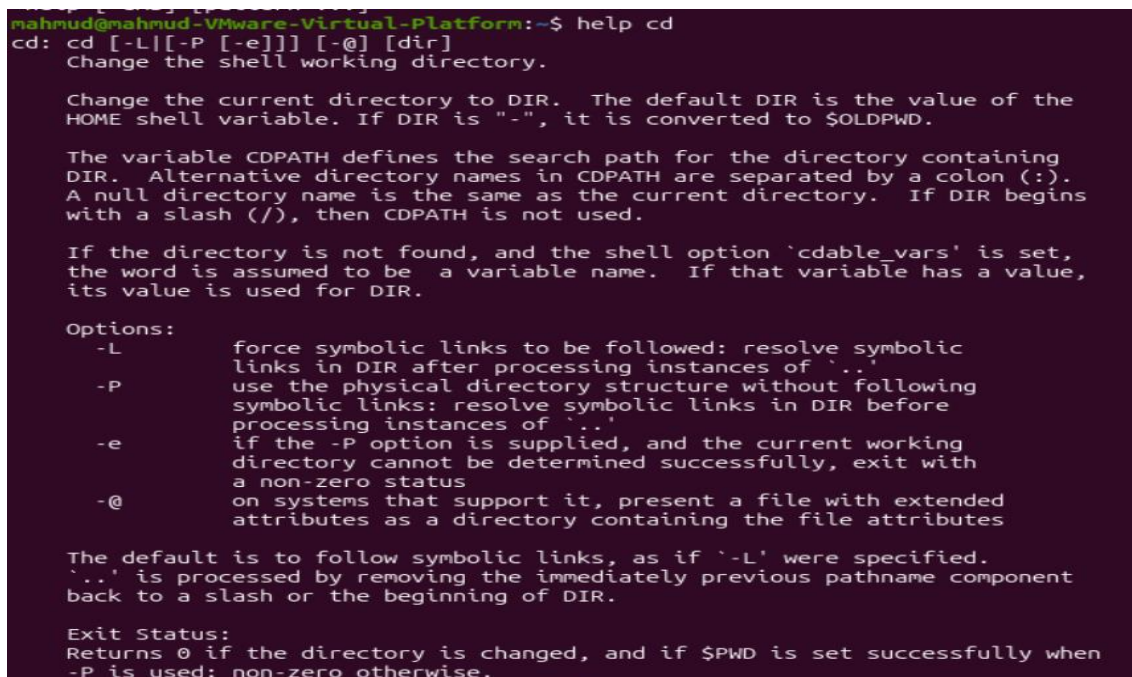
A star (*) next to a name means that the command is disabled.

! PIPELINE
job_spec [ & ]
(( expression ))
[ -p path ] Filename [ arguments ]
:
[ arg... ]
[[ expression ]]
alias [-p] [name=value] ...
bg [job_spec ...]
bind [-lpsvPSVK] [-n keymap] [-f filename] >
break [n]
builtin [shell-builtin [arg ...]]
caller [expr]
case WORD in [PATTERN] [PATTERN]...) COMMA-
cd [-L|-P [-e]] [-@] [dir]
command [-pOv] command [arg ...]
compgen [-V varname] [-abdefgjkuv] [-o op-
complete [-abdefgjkuv] [-pr] [-DEI] [-o o-
compsort [-o-o option] [-DEI] [name ...]
continue [n]
coproc [NAME] command [redirections]
declare [-aAffgIlnrtux] [name=value] ...>
dirs [-clv] [+N] [-n]
disown [-h] [-ar] [job_spec ...] pid ...]
echo [-neE] [arg ...]
enable [-a] [-dnps] [-f filename] [name ...]
eval [arg ...]
exec [-cl] [-a name] [command [argument ...]
exit [n]
export [-fn] [name=value] ...] or export ->
false
fc [-e ename] [-lnr] [first] [last] or fc ->
fg [job_spec]
for NAME [in WORDS ... ] ; do COMMANDS; don-
for (( exp1; exp2; exp3 )); do COMMANDS; do-
function name { COMMANDS ; } or name () { C-
getopts optstring name [arg ...]
hash [-lr] [-p pathname] [-dt] [name ...]
help [-dns] [pattern ...]
```

Command 2: help command_name

Description: Display detailed information about Linux command how it works.

Screenshot:



```
mahmud@mahmud-VMware-Virtual-Platform:~$ help cd
cd: cd [-L|-P [-e]] [-@] [dir]
Change the shell working directory.

Change the current directory to DIR. The default DIR is the value of the
HOME shell variable. If DIR is "-", it is converted to $OLDPWD.

The variable CDPATH defines the search path for the directory containing
DIR. Alternative directory names in CDPATH are separated by a colon (:).
A null directory name is the same as the current directory. If DIR begins
with a slash (/), then CDPATH is not used.

If the directory is not found, and the shell option 'cdable_vars' is set,
the word is assumed to be a variable name. If that variable has a value,
its value is used for DIR.

Options:
-L force symbolic links to be followed: resolve symbolic
-P use the physical directory structure without following
links in DIR after processing instances of '..'
-e if the -P option is supplied, and the current working
directory cannot be determined successfully, exit with
a non-zero status
-@ on systems that support it, present a file with extended
attributes as a directory containing the file attributes

The default is to follow symbolic links, as if '-L' were specified.
'..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.

Exit Status:
Returns 0 if the directory is changed, and if $PWD is set successfully when
-P is used; non-zero otherwise.
```

Command 3: pwd

Description: Display current directory

Screenshot:

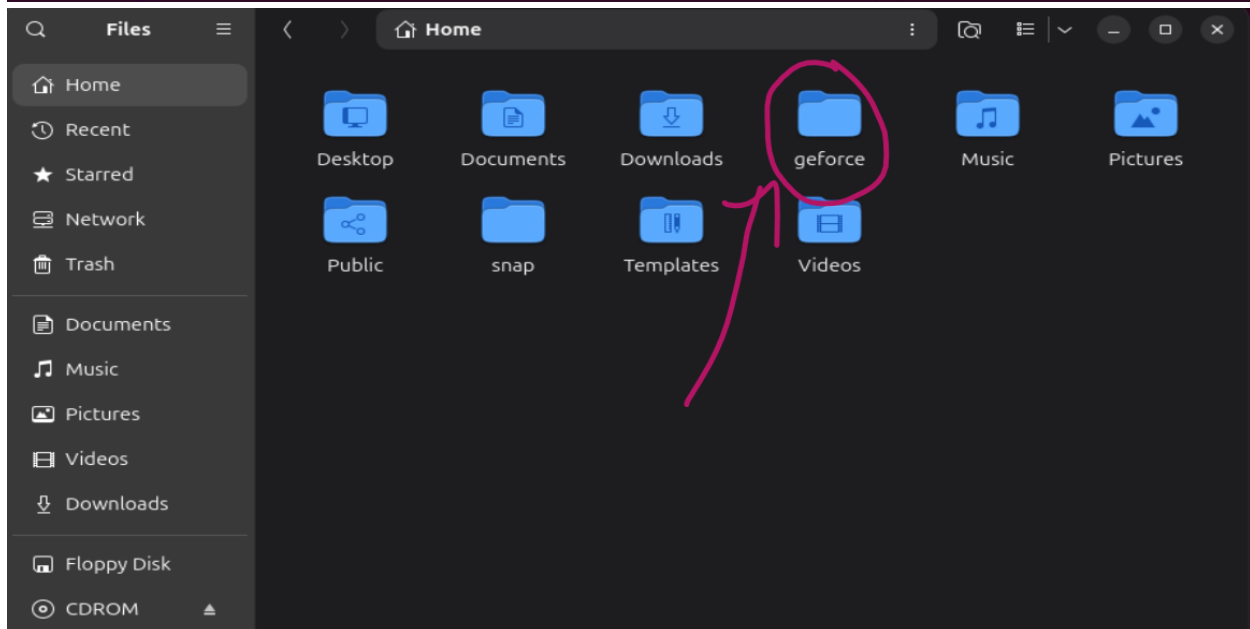
```
mahmud@mahmud-VMware-Virtual-Platform:~$ pwd
/home/mahmud
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 4: mkdir folder_name

Description: Create a new folder

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ mkdir geforce
mahmud@mahmud-VMware-Virtual-Platform:~$
```

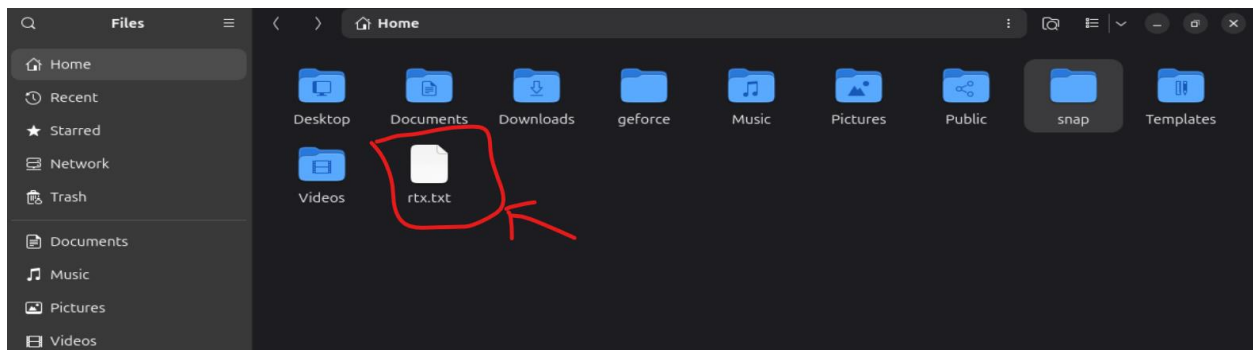


Command 5: touch file_name.file_extention

Description: Create a new file with particular format

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ touch rtx.txt
mahmud@mahmud-VMware-Virtual-Platform:~$
```



Command 6: `cd address`

Description: For change directory address

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ cd geforce
mahmud@mahmud-VMware-Virtual-Platform:~/geforce$ pwd
/home/mahmud/geforce
mahmud@mahmud-VMware-Virtual-Platform:~/geforce$
```

Command 7: `rm file_name`

Description: For remove file

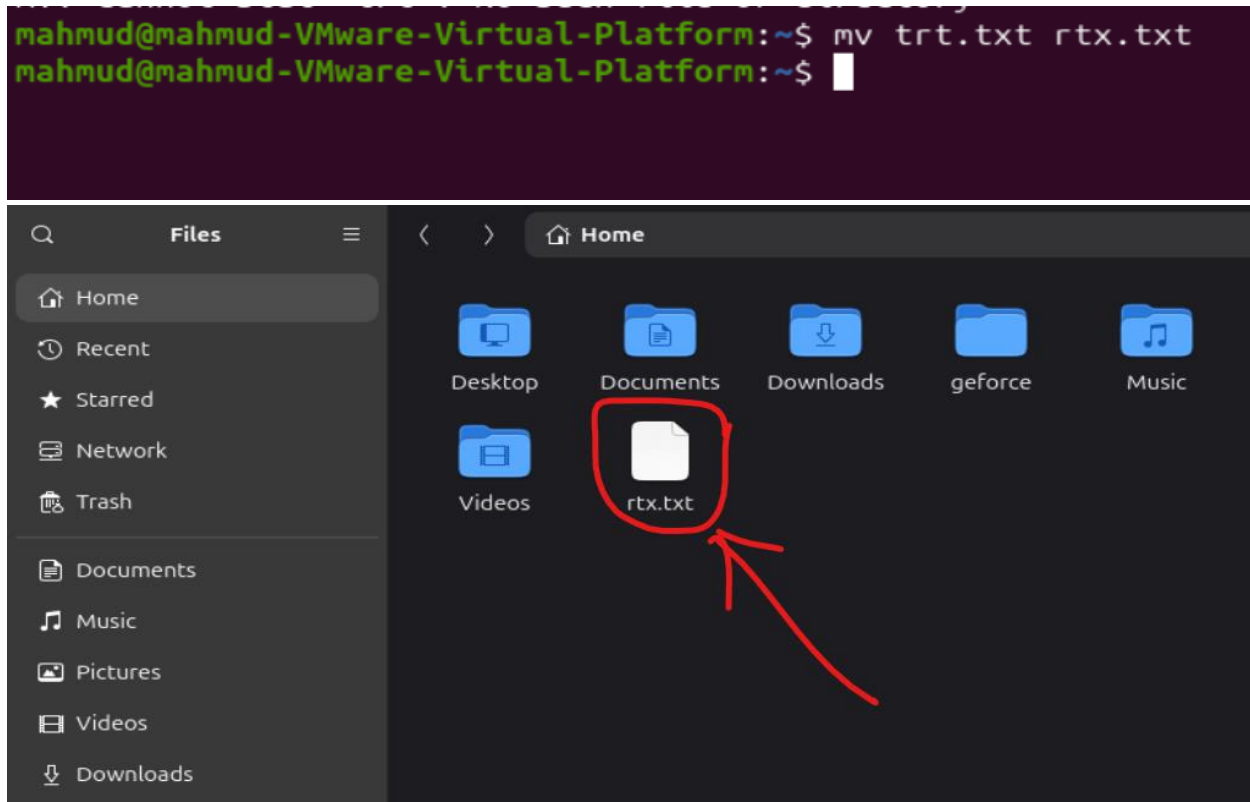
Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ rm rtx.txt
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 8: `mv file_previous_name file_updated_name`

Description: For rename file

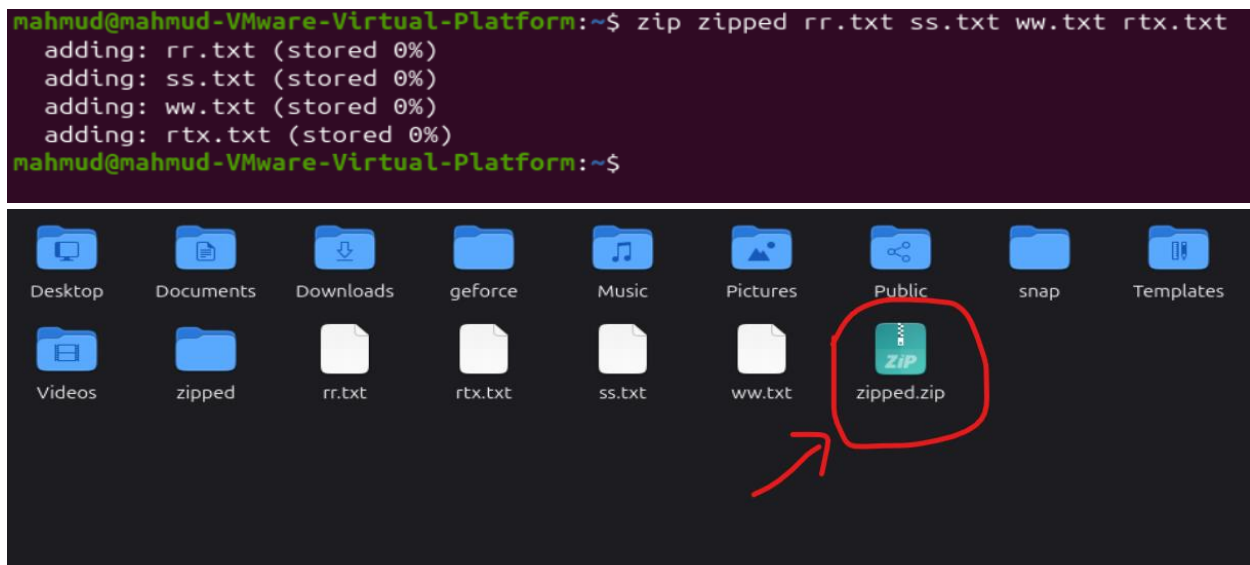
Screenshot:



Command 9: `zip zip_file_name file_name_1 file_name_2 .... file_name_n`

Description: For zip multiple file in on folder

Screenshot:



Command 10: unzip zip_file_name

Description: For unzip a zip file

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ unzip zipped
Archive:  zipped.zip
replace rr.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
extracting: rr.txt
replace ss.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
extracting: ss.txt
replace ww.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
extracting: ww.txt
replace rtx.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
extracting: rtx.txt
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 11: cat file_name1 file_name_2 --> file_name_3

Description: For combining multiple file info into 1 file

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ cat rr.txt ss.txt ww.txt --> rtx.txt
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 12: head -n number_of_line file_name.ext

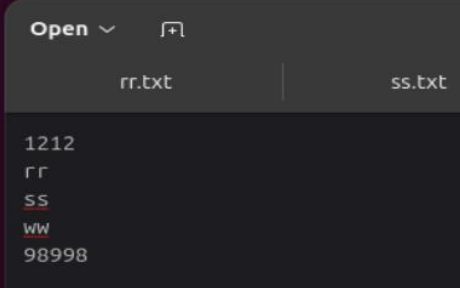
Description: to show first number_of_line

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ head -n
error: a value is required for '--lines <[.]NUM>' but none was supplied

For more information, try '--help'.
mahmud@mahmud-VMware-Virtual-Platform:~$ head -n rtx.txt
head: invalid number of lines: 'rtx.txt'
mahmud@mahmud-VMware-Virtual-Platform:~$ head -n 1 rtx.txt

1212
mahmud@mahmud-VMware-Virtual-Platform:~$
```



Command 13: `tail -n number_of_line file_name.ext`

Description: To show last number_of_line

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ tail -n 1 rtx.txt
98998
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 14: `grep keyword_name file_name.ext`

Description: To find the specific keyword in a file

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ grep rr rtx.txt
rr
mahmud@mahmud-VMware-Virtual-Platform:~$ grep 55 rtx.txt
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 15: `diff file_name_1 file_name_2`

Description: To find the difference between two fill. or find the similarity between two files

Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ diff rr.txt rtx.txt
0a1
> 1212
1a3,5
> 55
> WW
> 98998
mahmud@mahmud-VMware-Virtual-Platform:~$
```

Command 16: `wget url`

Description: To download a file form a specific url address.

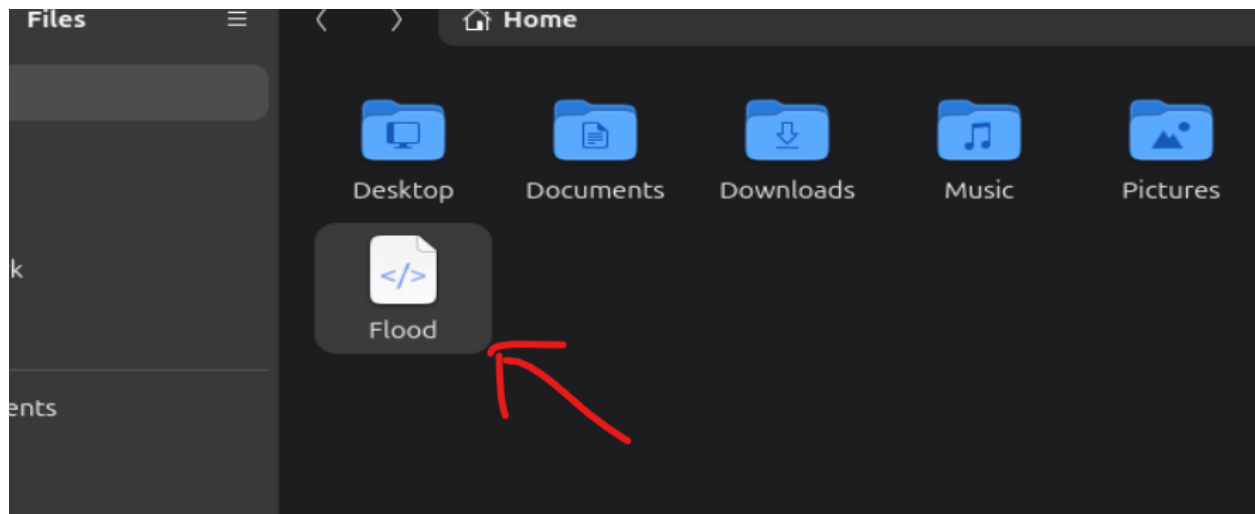
Screenshot:

```
mahmud@mahmud-VMware-Virtual-Platform:~$ wget https://en.wikipedia.org/wiki/Flood
--2026-07-13 22:15:06-- https://en.wikipedia.org/wiki/Flood
Resolving en.wikipedia.org (en.wikipedia.org)... 103.102.166.224, 2001:df2:e500:ed1a::1
Connecting to en.wikipedia.org (en.wikipedia.org)|103.102.166.224|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 567824 (555K) [text/html]
Saving to: 'Flood'

Flood                                100%[=====] 554.52K  2.37MB/s   in 0.2s

2026-07-13 22:15:06 (2.37 MB/s) - 'Flood' saved [567824/567824]

mahmud@mahmud-VMware-Virtual-Platform:~$
```



Conclusion:

In this experiment, fundamental Linux commands were practiced to understand file handling, navigate directory, searching text, archiving and remote file downloading. Gaining hands-on experience with these utilities highlights the efficiency of the Command Line Interface (CLI) over traditional graphical interfaces. Ultimately, mastering these core commands provides the essential foundation required to navigate and utilize the Linux operating system effectively in more advanced system-level tasks.